# COVID-19 Time-Series Analysis in R

*Analysing China's provincial COVID-19 confirmed-case series with correlation analysis and PCA, implemented entirely in R.*

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## Overview

This project analyses the Johns Hopkins CSSE global confirmed-cases dataset, focusing on China's 33 provinces over the period January to April 2020. Starting from cumulative case counts, the notebook derives daily increments, visualises provincial trends, quantifies inter-province correlations via Pearson heatmaps, and applies Principal Component Analysis to reduce the 72-day time series to a compact set of components.

## Goals

Reproduce a complete epidemiological time-series workflow in R, demonstrating that base R together with a small number of packages is sufficient for data wrangling, visualisation, correlation analysis, and dimensionality reduction on real-world public health data.

## Objectives

- Load the JHU CSSE dataset, isolate China's rows, and derive daily case counts from cumulative totals.
- Visualise daily confirmed cases for all provinces, with and without the Hubei outlier.
- Compute and display a Pearson correlation heatmap across all province pairs and rank the most correlated pairs.
- Apply PCA with prcomp, report explained and cumulative variance for the first three components, and scatter the data in PC space.
- Deliver the entire pipeline as a reproducible Jupyter notebook running an R kernel.

## Approach

Data wrangling is handled entirely in base R: row filtering, column dropping, matrix differencing for daily increments, and transposition to put provinces as columns. The pheatmap package renders the annotated correlation matrix. PCA is performed with prcomp (centred, unscaled) and the first three principal components are retained for visualisation using scatterplot3d and base R graphics.

## Outcomes

- The first principal component alone explains 99.9% of the variance, confirming that provincial outbreak trajectories are highly collinear.
- Cumulative variance reaches 99.97% with just three components, validating the low intrinsic dimensionality of the series.
- Province pairs such as Jiangsu-Anhui and Hunan-Guangdong reach Pearson correlations above 0.94, revealing strong geographic and temporal clustering.
- Hubei is confirmed as a structural outlier; removing it reveals clearer differentiation among the remaining provinces.